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PRACTICAL TRAINING: MEASURING TRAITS AND FITNESS COSTS OF EVOLVED POPULATIONS

(BACHELORSTUDIENGANG BIOLOGIE, 6. SEMESTER - MODUL 11-BIO-0630 – UMWELTMIKROBIOLOGIE)

1. OVERVIEW & SCHEDULE

This practical course aims to understand the importance of experimental evolution in studying adaptation of populations to biotic and abiotic factors. We will also mention coevolution of populations with their enemies to understand how interactions such as host-parasite and predator-prey evolve in time, and so affect the fitness of populations in different environments.

Microbial populations are challenged by biotic (e.g., predators and competitors) and abiotic (e.g., antibiotics and resource limitations) environments. Understanding how populations respond to these challenges via, for instance, standing genetic variation or *de novo* mutations is of great importance. Adaptations to novel environments come with fitness costs relative to the ancestral populations, which might be, for instance, a disadvantage in different environmental conditions. Since studies on experimental evolution need long-term lab work to reach many generations, we will use evolved bacterial populations from a previous experiment to measure intrinsic changes such as maximum growth rates and changes in functional traits such as resistance to predators or parasites, and also the fitness costs associated with these changes.

WEEK 1 **(4-8.5.2020)**

Day 1 ONLINE (Starting at 9:00)

Introduction to experimental evolution, coevolution, and fitness measurements. Supporting documents will be provided.

Day 2 LAB (8:00-12:00)

i) Incubation of evolved lines and ancestors.

ii) Preparation of experiments: Media, plates and phage dilutions.

Day 3 LAB (8:00-12:00)

i) Starting competition experiments take t_0 samples and plate.

ii) Starting growth rate measurements with/without phages.

Day 4 LAB (8:00-12:00)

i) Finishing competition experiments, plating t_1 samples.

ii) Counting t_0 samples.

Day 5 LAB (8:00-12:00)

i) Counting t_1 samples.

ii) Finishing growth rate measurements.

iii) Cleaning up.

WEEK 2 (11-15.5.2020)

Day 1 SELF STUDY & ONLINE (Starting at 9:00)

i) Data spreadsheets, organization.

ii) Question-answer session.

Day 2 ONLINE (Starting at 9:00)

i) R basics. Supporting documents will be provided.

ii) Analysing of competition experiments.

Day 3 ONLINE (Starting at 9:00)

i) R basics & Stats basics. Supporting codes will be provided.

ii) Analysing of growth measurements.

Day 4 ONLINE (Starting at 9:00)

i) R basics & Stats basics. Supporting codes will be provided.

ii) Analysis of growth curves.

Day 5 SELF STUDY & ONLINE (Starting at 9:00)

i) Analysis of phage resistance. Supporting codes will be provided.

ii) Wrapping up, question & answer.

2. BACKGROUND

Understanding how populations adjust in response to changing environmental conditions is of great importance. Shifting abiotic and biotic conditions might either cause maladaptation (Brady *et al.* 2019) or positive selection of populations (Kirkpatrick & Peischl 2013). Positive selection is possible with plasticity of physiological state of populations, standing genetic variation, or *de novo* mutations (Bleuven & Landry 2016). Every population has its own capacity, such as population size, growth rate, and mutation rate, to overcome novel conditions. Theoretical and experimental studies have shown that evolutionary adaptation depends on a variety of factors such as population size (Bell & Gonzalez 2009), genotypic and species diversity (Osmond & de Mazancourt 2012; Ramsayer *et al.* 2013; Low-Décarie *et al.* 2015), rate of environmental change (Bell & Gonzalez 2011; Toprak *et al.* 2012), environmental fluctuations (Bell & Gonzalez 2011), the strength of selection, and history of stress (Samani & Bell 2018). However, the rate of environmental shift and biotic interactions such as host-pathogen interactions can alter the response of populations (Bell 2017). It has been shown that host and pathogen traits may change over time due to the trade-offs between resistance and growth (Gomez & Buckling 2011a; Hall *et al.* 2011; Lopez Pascua *et al.* 2012; Koskella & Brockhurst 2014; Lopez Pascua *et al.* 2014; Frickel *et al.* 2016).

Understanding how populations develop resistance is important both to understand ecological interactions and to better predict environmental and medical applications. One example is the application of phage therapy against bacterial infections. Since there is a global rise in resistance to antibiotics, there has been a renewed interest in the use of phages to treat bacterial infections (Walker *et al.* 2017). However, bacterial resistance to phages could limit the efficacy of long-term treatments (Rohde *et al.* 2018). Therefore, understanding how bacterial populations develop phage resistance is of great importance. For example, a common form of phage resistance is to limit adsorption by modifying the targeted surface molecule(s) that bind phage receptor(s). Since these receptors have also other functions, such as motility and nutrient transport, resistance is often associated with fitness costs, which might be a disadvantage in the absence of phages (Scanlan *et al.* 2015a). It has been shown that

resistance evolution might limit the beneficial mutations that improve the fitness of bacteria in different environments (Scanlan *et al.* 2015b). Note that antagonistic coevolution (e.g., reciprocal selection for host resistance and parasite infectivity) might cause phenotypes of host and parasite change over time. Coevolution between bacteria and phages has been shown in various natural ecosystems (e.g. Friman *et al.* 2011), as well as in laboratory microcosms (e.g. Gomez & Buckling 2011b).

Controlled laboratory populations are often employed to study the mechanisms of evolution. Predictions of evolutionary theory can be tested and new outcomes might serve to develop the evolutionary ecology theory (Van den Bergh *et al.* 2018; McDonald 2019). Since experimental evolution allow testing, for instance, different bacteria-phage combinations and experimental environments, also the other phenotypic effects and costs in terms of growth rates and competitive abilities, factors which determine the infectivity of phages can be defined (Scanlan *et al.* 2015a).

3. METHODS

i) Organisms

We have performed an experimental evolution using *Escherichia coli* strain B REL607. Strains from different microcosm setups were isolated and stored at -80°C. A subset of microcosms contained lytic bacteriophage T4 and allowed to coevolve with this host bacterial strain. We also have kept an isogenic mutant, *E. coli* B REL606, of this bacterial strain at -80°C as nonevolved ancestral. *E. coli* REL607 and *E. coli* REL606 can be differentiated with Ara+ and Ara- genomic markers, respectively. Ara+ and Ara- cells produce red and pink colonies, respectively, on Tetrazolium Arabinose (TA) agar plates (See additional information below), which help us to differentiate ancestral and evolved lines on plates.

ii) Competition experiments and measuring relative fitness

We will recover the clones and compete ancestral and evolved lines together for 24 h to calculate fitness cost as follows:

Figure 1: Typical fitness measurements, shown as Figure 4 in McDonald 2019 (not reproduced here).

1. One *E. coli* clone from three samples of incubations without phage and three samples of incubations with phage will be randomly selected. We will recover these clones after incubation overnight at 30°C in 25 ml bottles containing 6 mL LB medium. We will also grow one ancestral clone under the same conditions.
2. The next day, every clone will be diluted 100-fold (~10^7 cells ml^{-1}) and mixed 1:1 (v/v) with the ancestral clone. These mixtures will be plated on TA (tetrazolium arabinose) agar with appropriate dilutions to determine the evolved and ancestral abundance at t0. These mixtures will be grown 24h at 30°C (Figure 1).
3. After 24 h, cultures will be plated on TA agar with dilutions to estimate the relative abundances of ancestral and evolved clones at t1. Colony-forming units (CFU) of t_0 cultures will be counted on this day after ~24 h.
4. Colony-forming units (CFU) of t_1 cultures will be counted after 24 hours again. Fitness cost can be calculated using the following formula:

$$s = \frac{\ln\left[\frac{N_E(t_1)}{N_A(t_1)}\right] - \ln\left[\frac{N_E(t_0)}{N_A(t_0)}\right]}{t} \quad \text{(eq. 1)}$$

where s is the selection coefficient, and s>1 indicates higher fitness of evolved populations relative to the ancestor. t is the time interval (1 day); t_0 is the start and t1 is the end of the experiment. N_E is the abundance of evolved, and N_A is the abundance of ancestral populations.

iii) Measuring absolute fitness

Growth rates will be used to estimate focal species' absolute fitness across the microcosms. 1% of the recovered clones will be transferred into fresh LB media, and the OD_{600nm} will be measured at 15-minute intervals in an automated microplate at

30°C for 48 h (Figure 2). The measurements will be used to fit a logistic growth model:

$$N_t = KN_0e^{rt}/K + N_0(e^{rt} - 1) \quad \text{(eq. 2)}$$

where N_t is the observed population size at time *t.* Estimated parameters are initial population size N_0, carrying capacity *K,* and growth rate *r.*

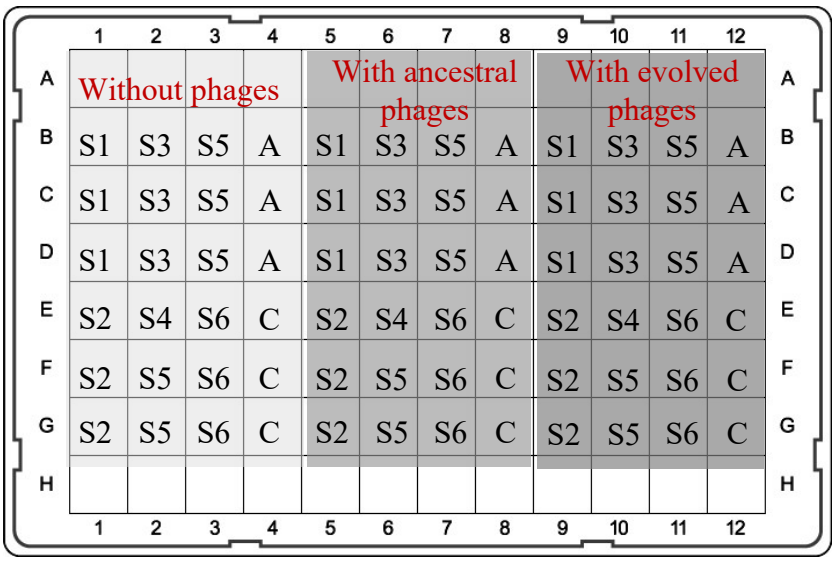


Figure 2 Typical growth rate/resistance co-testing setup. S represents the samples, A is ancestral population, C is sterile control. Optionally coevolved phages can be also tested for virulence.

iv) Measuring phage resistance

Phage resistance will be estimated using the growth curves of clones that were incubated with phages. 1% of the recovered clones will be transferred into fresh LB media and supplemented with 10^7 phage particles, and the OD600nm will be measured at 15-minute intervals in an automated microplate at 30°C for 48 h. Additionally, the virulence of the phage T4 will be determined using the growth curves of clones in the presence of T4 phages (Figure 2). Resistance will be estimated using a

liquid assay score by calculating the area under the curve for both growth curves with and without phages:

$$area\ under\ the\ curve = \sum_{i=1}^{24} \frac{OD_{i+1} + OD_i}{2} \quad \text{(eq. 3)}$$

$$liquid\ assay\ score = \frac{Area_{positiv\ control} - Area_{phage\ treatment}}{Area_{positiv\ control}} \cdot 100 \quad \text{(eq. 4)}$$

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MEDIA RECIPES

Tetrazolium Arabinose plates

Tryptone	10 g
Yeast extract	1 g
Sodium chloride	5 g
Agar	16 g
Antifoam (5%)	1 ml
dH_20	1000 ml
L (+) Arabinose	10 g
TTC (5%)	1 ml (TTC is the tetrazolium indicator dye)

Split the water, and autoclave the arabinose separately from the rest of the medium. Sterilize by autoclaving for 20 min at 15 psi (1.05 kg/cm^2) on liquid cycle. Combine the two parts after autoclaving, and then add the TTC from a sterile stock solution and mix.

Source: Lenski Lab, http://myxo.css.msu.edu/ecoli/taagar.html

LB Medium

Tryptone	10 g
Yeast extract	1 g
Sodium chloride	5 g
dH_2O	1000 ml

Combine the reagents and shake until the solutes have dissolved. Adjust the pH to 7.0 with 5 N NaOH (~0.2 mL). Adjust the final volume of the solution to 1 L with H_2O. Sterilize by autoclaving for 20 min at 15 psi (1.05 kg/cm^2) on liquid cycle.

Source: CSH Protocols, http://cshprotocols.cshlp.org/content/2006/1/pdb.rec8247

PBS Buffer

Reagent	**Amount to add (for 1× solution)**
NaCl	8 g
KCl	0.2 g
Na_2HPO_4	1.44 g
KH_2PO_4	0.24 g
$CaCl_2 \cdot 2H_2O$	0.133 g
$MgCl_2 \cdot 6H_2O$	0.10 g

PBS can be made as a 1× solution or as a 10× stock. To prepare 1 L of either 1× or 10× PBS, dissolve the reagents listed above in 800 mL of H_2O. Adjust the pH to 7.4 (or 7.2, if required) with HCl, and then add H_2O to 1 L. Dispense the solution into aliquots and sterilize them by autoclaving for 20 min at 15 psi (1.05 kg/cm^2) on liquid cycle or by filter sterilization. Store PBS at room temperature.

Source: CSH Protocols, http://cshprotocols.cshlp.org/content/2006/1/pdb.rec8247